

0.1 Section 4.4 - Linear Congruences

To think of linear congruences, let's start with an example

Example 1

A group of soldiers line up in rows of 8. However, there are 4 soldiers left over. When the soldiers line up in rows of 7, there are 5 soldiers left over. What is the least possible number of soldiers in the group?

Let x = number of soldiers

From the description, we know:

$$x \equiv 4 \pmod{8}$$

$$x \equiv 5 \pmod{7}$$

This also means that

$$\exists k_1 \in \mathbb{Z} \text{ such that } x = 8k_1 + 4$$

$$8k_1 + 4 \equiv 5 \pmod{7}$$

(substitute x)

$$8k_1 \equiv 1 \pmod{7}$$

$$k_1 \equiv 1 \pmod{7}$$

(since 8^{-1} is 1)

Expanding that last equation, we get

$$\exists k_2 \in \mathbb{Z}, k_1 = 7k_2 + 1$$

Substituting k_1 back into the equation for x , we get

$$x = 8k_1 + 4 \quad \text{(original)}$$

$$x = 8(7k_2 + 1) + 4 \quad \text{(substitute)}$$

$$x = 56k_2 + 12$$

Since we're looking for the smallest number of soldiers, we can set $k_2 = 0$ and we get $x = 12$ as our final answer

This is actually an example of the **Chinese Remainder Theorem**

Definition 1 (Relatively Prime)

Two numbers $a, b \in \mathbb{Z}$ are relatively prime if

$$\gcd(a, b) = 1$$

Theorem 1 (Chinese Remainder Theorem)

Basically

Let $m_1, m_2, \dots, m_n \in \mathbb{Z}^+$ which are pairwise relatively prime (they're all relatively prime to each other) and $a_1, a_2, \dots, a_n \in \mathbb{Z}$, then the system

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\vdots \\x &\equiv a_n \pmod{m_n}\end{aligned}$$

Has a unique solution that's mod $M = m_1 \cdot m_2 \cdot \dots \cdot m_n$

Example 2

Let's start with the system

$$\begin{aligned}x &\equiv 14 \pmod{16} \\x &\equiv 21 \pmod{31} \\x &\equiv 12 \pmod{41}\end{aligned}$$

We start by rewriting the first equation

$$x = 16k_1 + 14$$

And substituting it into the second equation

$$16k_1 + 14 \equiv 21 \pmod{31}$$

Now isolate k_1 . You need to find 16^{-1} , but thankfully we have some shortcuts

$$\begin{aligned}
16k_1 &\equiv 7 \pmod{31} \\
32k_1 &\equiv 14 \pmod{31} && \text{(multiply 2)} \\
k_1 &\equiv 14 \pmod{31} && (32 \equiv 1 \pmod{31})
\end{aligned}$$

Now that we have k_1 , let's expand it

$$k_1 = 31k_2 + 14$$

And plug it back into our equation of x

$$\begin{aligned}
x &= 16k_1 + 14 && \text{(original)} \\
x &= 16(31k_2 + 14) + 14 \\
x &= 16 \cdot 31k_2 + 16 \cdot 14 + 14 \\
\boxed{x} &= \boxed{496k_2 + 238}
\end{aligned}$$

Now that we have a new equation for x involving k_2 , let's use our last equation

$$\begin{aligned}
x &\equiv 12 \pmod{41} && \text{(last equation)} \\
496k_2 + 238 &\equiv 12 \pmod{41} \\
496k_2 &\equiv -226 \pmod{41} \\
4k_2 &\equiv 20 \pmod{41} && \text{(reduce 496 and -226)}
\end{aligned}$$

Now we gotta find $4^{-1} \pmod{41}$

Let's use the Euclidean Algorithm

$$\begin{aligned}
41 &= 10 \cdot 4 + 1 \\
1 &= 41 - 10 \cdot 4 \\
4^{-1} &\equiv -10 \pmod{41} \\
\boxed{4^{-1}} &= \boxed{31 \pmod{41}} && \text{(add 41)}
\end{aligned}$$

Now that we have 4^{-1} , we solve

$$\begin{aligned}
 4k_2 &\equiv 20 \pmod{41} \\
 4^{-1} \cdot 4k_2 &\equiv 4^{-1} \cdot 20 \pmod{41} \\
 31 \cdot 4k_2 &\equiv 31 \cdot 20 \pmod{41} \\
 k_2 &\equiv 620 \pmod{41} \\
 \boxed{k_2 &\equiv 5 \pmod{41}}
 \end{aligned}$$

Now that we solved for k_2 , let's expand it one last time

$$k_2 = 41k_3 + 5$$

And plug it back into our equation for x involving k_2

$$\begin{aligned}
 x &= 496k_2 + 238 && \text{(original)} \\
 x &= 496(41k_3 + 5) + 238 && \text{(substituted)} \\
 \boxed{x &= 20336k_3 + 2718} && \text{(simply)}
 \end{aligned}$$

Now that we simplified for the equation, we see that it represents

$$\boxed{x \equiv 2718 \pmod{20336}}$$

And you can see from the Chinese Remainder Theorem that our mod is actually the product of our moduli

$$M = 20336 = 16 \cdot 31 \cdot 41$$

That was... very long, so let's write out the steps:

0.1.1 Steps for Solving Linear Congruences

Given the system: (I'm just gonna use up to 3 for this)

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\x &\equiv a_3 \pmod{m_3}\end{aligned}$$

The system has a **unique solution** if m_1, m_2, m_3 are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1 k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$\begin{aligned}x &\equiv a_2 \pmod{m_2} \\m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2}\end{aligned}$$

And solve for k_1

$$\begin{aligned}m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\m_1^{-1} k_1 &\equiv m_1^{-1} (a_2 - a_1) \pmod{m_2} \\k_1 &\equiv m_1^{-1} (a_2 - a_1) \pmod{m_2} \\ \boxed{k_1 \equiv c_1} &\pmod{m_2} \qquad (c_1 \text{ is shorthand})\end{aligned}$$

Note

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of m_1

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, k_2 \in \mathbb{Z}$$

3. Plug that into our equation for x

$$x = m_1 k_1 + a_1$$

$$\boxed{x = m_1(m_2 k_2 + c_1) + a_1}$$

Plug that into the last equation

$$m_1(m_2 k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for $k_2 \pmod{m_3}$

$\vdots$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3 k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for x

$$\begin{aligned}
 x &= m_1(m_2k_2 + c_1) + a_1 \\
 x &= m_1(m_2(m_3k_3 + c_2) + c_1) + a_1 \\
 \boxed{x &= m_1m_2m_3k_3 + c_3}
 \end{aligned}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3} \pmod{m_1m_2m_3}$$

Basically, starting with the first equation

1. Expand first, solving for x
2. Plug x into second
3. Solve for k_1
4. Expand k_1
5. Plug k_1 back into first expanded of x (one with k_1)
6. Solve for k_2
7. Expand k_2
8. Plug k_2 back into second expanded of x (one with k_2)
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation

Let's do something else now

0.1.2 Factorials

1. $1! \pmod{2} = 1$
2. $2! \pmod{3} = 2$
3. $3! \pmod{4} = 2$
4. $4! \pmod{5} = 4$
5. $5! \pmod{6} = 0$, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{6} = 0$

$$6. \ 6! \pmod{7} = 6$$

$$7. \ 7! \pmod{8} = 0, \ 7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{8} = 0$$

$$8. \ 8! \pmod{9} = 0, \ 8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{9} = 0$$

$$9. \ 9! \pmod{10} = 0, \ 9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{10} = 0$$

$$10. \ 10! \pmod{11} = 10$$

$$11. \ 999! \pmod{1000} = 0 \text{ since } 1000 \text{ is composite}$$

Notice that for all the ones that are not 0, the modulus is prime (except 4)

This brings us to our next theorems

0.1.3 Wilson's Theorem

Theorem 2 (Wilson's Theorem)

If p is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

Theorem 3 (Some Unnamed Theorem)

If n is a composite integer greater than 4,

then

$$(n-1)! \equiv 0 \pmod{n}$$

If $n = 4$, then

$$3! \equiv 2 \pmod{4}$$

0.1.4 Fermat's Little Theorem

There's examples that derive this, but I'm too lazy

Theorem 4 (Fermat's Little Theorem)

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof

Consider $S = \{1, 2, \dots, p-1\}, p \in \mathbb{Z}$

Let's define a function $f : S \rightarrow S, f(x) = ax \pmod{p}$

Example 3

Let $p = 7, a = 3$

Then $S = \{1, 2, 3, 4, 5, 6\}$ and $f(x) = 3x \pmod{7}$

x	$f(x)$
1	3
2	6
3	2
4	5
5	1
6	4

We see that f is a **bijection**

so we claim f is a **bijection**

Why?

Let $y \in S$

We must solve

$$ax \equiv y \pmod{p}$$

and since $\gcd(a, p) = 1$, then a^{-1} exists $\pmod{p}$

Therefore, $a^{-1}(ax) \equiv a^{-1}y \pmod{p}$

or

$$x \equiv a^{-1}y \pmod{p}$$

we conclude that f is **onto (surjective)**

For the injective part, suppose $f(x_1) = f(x_2)$

Then

$$ax_1 \equiv ax_2 \pmod{p}$$

Since

$$\gcd(a, p) = 1, \text{ then } a^{-1} \text{ exists } \pmod{p}$$

Therefore

$$a^{-1}(ax_1) \equiv a^{-1}(ax_2) \pmod{p}$$

or

$$x_1 \equiv x_2 \pmod{p}$$

We conclude f is **one to one (injective)**

Now let's consider the product $f(1) \cdot f(2) \cdot \dots \cdot f(p-1)$

$$f(1) \cdot f(2) \cdot \dots \cdot f(p-1) \equiv 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \pmod{p}$$

$$a \cdot 2a \cdot \dots \cdot (p-1)a \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

$$a^{p-1} [1 \cdot 2 \cdot \dots \cdot (p-1)] \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

By Wilson's Theorem we get

$$(p-1)! \equiv p-1 \pmod{p}$$

$$a^{p-1}! \equiv (p-1)! \pmod{p}$$

$$a^{p-1}(p-1) \equiv p-1 \pmod{p}$$

$$a^{p-1} \equiv 1 \pmod{p} \text{ as desired}$$

Let's do some examples

Example 4

Compute the following

$$9^{2026} \pmod{11}$$

$$a : 9$$

$$p : 11$$

$$\gcd(9, 11) = 1$$

By Fermat's Little Theorem,

$$9^{10} \equiv 1 \pmod{11}$$

We can use this to break up 2026

$$\begin{aligned} 9^{2026} &\equiv 9^{202 \cdot 10 + 6} \pmod{11} \\ &\equiv 9^{202 \cdot 10} \cdot 9^6 \pmod{11} \\ &\equiv 9^6 \pmod{11} \end{aligned}$$

Remember that we can subtract 11 from 9, so $9 \equiv -2 \pmod{11}$

This makes

$$\begin{aligned} &\equiv 9^6 \pmod{11} \\ &\equiv (-2)^6 \pmod{11} \\ &\equiv 64 \pmod{11} \\ &\equiv 9 \pmod{11} \end{aligned}$$

0.1.5 Extended Fermat's Little Theorem

There's also an extension of this

Theorem 5 (Extended Fermat's Little Theorem)

If $a, p, q \in \mathbb{Z}$ with

1. p and q are distinct primes
2. $\gcd(a, pq) = 1$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

Example 5

Compute

1.

$$13^{2026} \pmod{253}$$

First, let's check:

(a) $253 = 11 \cdot 23$

$p = 11, q = 23$, are distinct primes

(b) $\gcd(13, 253) = 1$

This means we can use Extended Fermat's Little Theorem (EFLT)

$$13^{(11-1)(23-1)} \equiv 1 \pmod{253}$$

$$13^{10 \cdot 22} \equiv 1 \pmod{253}$$

$$13^{220} \equiv 1 \pmod{253}$$

Breaking up 2026,

$$2026 = 9 \cdot 220 + 46 \pmod{253}$$

This means

$$13^{2026} \equiv 13^{46} \pmod{253}$$

$$13 \pmod{253} = 13$$

$$\begin{aligned} 13^2 \pmod{253} &= 13^2 \\ &= 169 \end{aligned}$$

$$\begin{aligned} 13^4 \pmod{253} &= 169^2 \pmod{253} \\ &= 225 \end{aligned}$$

$$\begin{aligned} 13^8 \pmod{253} &= 225^2 \pmod{253} \\ &= 25 \end{aligned}$$

$$\begin{aligned} 13^{16} \pmod{253} &= 25^2 \pmod{253} \\ &= 119 \end{aligned}$$

$$\begin{aligned} 13^{32} \pmod{253} &= 119^2 \pmod{253} \\ &= 246 \end{aligned}$$

Now let's break up 46 into powers of 2

$$\begin{aligned}
13^{2026} &\equiv 13^{46} \\
&\equiv 13^{32} \cdot 13^8 \cdot 13^4 \cdot 13^2 \pmod{253} \\
&\equiv 246 \cdot 25 \cdot 225 \cdot 169 \pmod{253} \\
&\equiv 31
\end{aligned}$$

2.

$$24^{2026} \pmod{31}$$

(a) $p = 31$ is prime

(b) $\gcd(24, 31) = 1$

so FLT applies

$$\begin{aligned}
24^{31-1} &\equiv 1 \pmod{31} \\
24^{30} &\equiv 1 \pmod{31}
\end{aligned}$$

$$2026 = 67 \cdot 30 + 16$$

$$\begin{aligned}
24^{2026} &\equiv 24^{67 \cdot 30 + 16} \pmod{31} \\
&\equiv (24^{30})^{67} \cdot 24^{16} \pmod{31} \\
&\equiv 1^{67} \cdot 24^{16} \pmod{31} \\
&\equiv 24^{16} \pmod{31} \\
&\equiv 7 \pmod{31} \qquad \text{(brute forced)}
\end{aligned}$$

0.1.6 An Implication of Fermat's Little Theorem

Definition 2 (Implication of FLT)

I'm not sure if this is called something but recall that Fermat's Little theorem states that

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

But actually, any multiple of $p - 1$ will make a equivalent to $1 \pmod{p}$

We can rewrite this as

$$a^{k(p-1)} \equiv 1 \pmod{p}, k \in \mathbb{Z}^+$$

But this another way of saying this is

$$a^k \equiv 1 \pmod{p}, k \equiv 0 \pmod{p-1}$$

The second half just means that k is a multiple of $p - 1$

So TDLR, if a is raised to a multiple of $p - 1$, then it is equivalent to $1 \pmod{p}$

This also implies that

$$a^x \equiv a^y \pmod{p} \text{ if } x \equiv y \pmod{p-1}$$

We'll see this being used later